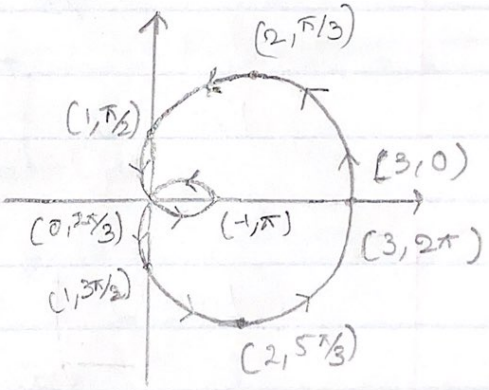
Final Exam Take Home:

* I Agree

① $r = 2\cos\theta + 1$

(A)

θ	$r = 2\cos\theta + 1$
0	3
$\pi/3$	2
$\pi/2$	1
π	-1
$2\pi/3$	0
$3\pi/2$	1
$5\pi/3$	2
2π	3



(B)

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

$a = 2\pi/3$ and $b = \pi$, when $L = 2\int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$

$$\therefore \quad L = 2\int_{2\pi/3}^{\pi} \sqrt{(2\cos\theta + 1)^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

$r = 2\cos\theta + 1 \quad , \quad \frac{dr}{d\theta} = -2\sin\theta$

$$\left(\frac{dr}{d\theta}\right)^2 = (-2\sin\theta)^2 = 4\sin^2\theta$$

$$\therefore \quad L = 2\int_{2\pi/3}^{\pi} \sqrt{(2\cos\theta+1)^2 + 4\sin^2\theta} \; d\theta$$

$$L = 2\int_{2\pi/3}^{\pi} \sqrt{4\cos^2\theta + 4\cos\theta + 1 + 4\sin^2\theta} \; d\theta$$

$$L = 2\int_{2\pi/3}^{\pi} \sqrt{4\cos\theta + 1 + 4} \; d\theta$$

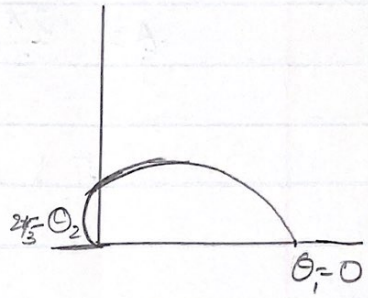
$$\boxed{L = 2\int_{2\pi/3}^{\pi} \sqrt{4\cos\theta + 5} \; d\theta}$$

(c)

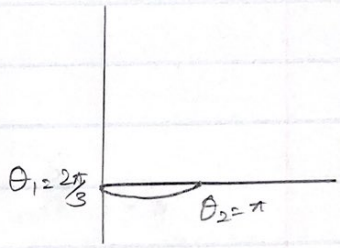
$$A = \frac{1}{2}\int_{\theta_1}^{\theta_2} r^2 \, d\theta \quad (\text{or})$$

$$A = \int_a^b \frac{1}{2}[f(\theta)]^2 \, d\theta - \int_c^b \frac{1}{2}[g(\theta)]^2 \, d\theta$$

$A =$ outer loop $-$ inner loop.



$$A = 2\int_0^{2\pi/3} \left(\frac{1}{2}\right)(1+2\cos\theta)^2\, d\theta - 2\int_{\frac{2\pi}{3}}^{\pi} \left(\frac{1}{2}\right)(1+2\cos\theta)^2\, d\theta$$



$$A = \int_0^{2\pi/3} (1+2\cos\theta)^2\, d\theta - \int_{2\pi/3}^{\pi} (1+2\cos\theta)^2\, d\theta$$

$$A = \int_0^{2\pi/3} (1+4\cos\theta+4\cos^2\theta)\, d\theta - \int_{2\pi/3}^{\pi} (1+4\cos\theta+4\cos^2\theta)\, d\theta$$

$$A = \int_0^{2\pi/3} (1)\, d\theta + \int_0^{2\pi/3} 4\cos\theta\, d\theta + \int_0^{2\pi/3} 4\cos^2\theta\, d\theta - \int_{2\pi/3}^{\pi} (1)\, d\theta - \int_{2\pi/3}^{\pi} 4\cos\theta\, d\theta - \int_{2\pi/3}^{\pi} 4\cos^2\theta\, d\theta$$

$$A = \left[\frac{2\pi}{3} - 0\right] - \left[\pi - \frac{2\pi}{3}\right] + 4\left[\sin\frac{2\pi}{3} - \sin 0\right] - 4\left[\sin\pi - \sin\frac{2\pi}{3}\right] + \int_0^{2\pi/3} 2(1+\cos 2\theta)\, d\theta - \int_{2\pi/3}^{\pi} 2(1+\cos 2\theta)\, d\theta$$

$$A = \frac{2\pi}{3} - \frac{\pi}{3} + 4\left[\frac{\sqrt{3}}{2} - 0\right] - 4\left[0 - \frac{\sqrt{3}}{2}\right] + \int_0^{2\pi/3} 2\, d\theta + \int_0^{2\pi/3} 2\cos(2\theta)\, d\theta - \int_{2\pi/3}^{\pi} 2\, d\theta - \int_{2\pi/3}^{\pi} 2\cos(2\theta)\, d\theta$$

$$A = \frac{\pi}{3} + 2\sqrt{3} + 2\sqrt{3} + \left[\frac{4\pi}{3} - 0\right] + \left[\sin 4\pi/3 - \sin 0\right] - \left[2\pi - \frac{4\pi}{3}\right] - \left[\sin 2\pi - \sin\frac{4\pi}{3}\right]$$

$$A = \frac{\pi}{3} + \frac{4\pi}{3} + 4\sqrt{3} + \left(-\frac{\sqrt{3}}{2}\right) - \frac{2\pi}{3} + \left(-\frac{\sqrt{3}}{2}\right)$$

$$A = \frac{5\pi}{3} - \frac{2\pi}{3} + 4\sqrt{3} - \sqrt{3}$$

$$A = \frac{3\pi}{3} + 3\sqrt{3} \Rightarrow A = \pi + 3\sqrt{3}$$

∴ Area of the region between inner and outer loop of the limacon $= \pi + 3\sqrt{3}$

(2) $r = 1 - \cos\theta$

(A) $\theta = \pi/6$

$$\frac{dy}{dx} = \frac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$$

$$r' = \sin\theta \quad , \quad r = 1-\cos\theta \quad , \quad \theta = \pi/6$$

$$\frac{dy}{dx} = \frac{\sin\theta\,\sin\theta + (1-\cos\theta)\cos\theta}{\sin\theta\,\cos\theta - (1-\cos\theta)\sin\theta}\Bigg|_{\theta=\pi/6}$$

$$\frac{dy}{dx} = \frac{\sin^2\theta + \cos\theta - \cos^2\theta}{\sin\theta\cos\theta - \sin\theta + \sin\theta\cos\theta}\Bigg|_{\theta=\pi/6}$$

$$\left.\frac{dy}{dx}\right|_{\theta=\pi/6} = \frac{\left(\sin(\pi/6)\right)^2 + \cos(\pi/6) - \left(\cos(\pi/6)\right)^2}{2\left(\sin(\pi/6)\cos(\pi/6)\right) - \sin(\pi/6)}$$

$$\left.\frac{dy}{dx}\right|_{\theta=\pi/6} = \frac{(1/2)^2 + \frac{\sqrt{3}}{2} - \left(\frac{\sqrt{3}}{2}\right)^2}{2\left((1/2)(\sqrt{3}/2)\right) - 1/2} = \frac{\frac{1}{4} - \frac{3}{4} + \frac{\sqrt{3}}{2}}{\sqrt{3}/2 - 1/2}$$

$$= \frac{\frac{\sqrt{3}}{2} - \frac{1}{2}}{\sqrt{3}/2 - 1/2} = 1$$

$$y - y_1 = 1\,(x - x_1)$$

$$x_1 = r\cos\theta\Big|_{\theta=\pi/6} \Rightarrow x_1 = (1-\cos\theta)(\cos\theta\Big|_{\theta=\pi/6}$$

$$\Rightarrow x_1 = \left(1 - \frac{\sqrt{3}}{2}\right)\frac{\sqrt{3}}{2}$$

$$\Rightarrow x_1 = \frac{\sqrt{3}}{2} - \frac{3}{4} = \frac{2\sqrt{3}-3}{4}$$

$$y_1 = r\sin\theta\Big|_{\theta=\pi/6}$$

$$\Rightarrow y_1 = (1-\cos\theta)\sin\theta\Big|_{\theta=\pi/6}$$

$$y_1 = \left(1 - \frac{\sqrt{3}}{2}\right)\frac{1}{2} = \frac{1}{2} - \frac{\sqrt{3}}{4} = \left(\frac{2-\sqrt{3}}{4}\right)$$

$$\Rightarrow y - \left(\frac{2-\sqrt{3}}{4}\right) = 1\left(x - \left(\frac{2\sqrt{3}-3}{4}\right)\right)$$

$$\Rightarrow 4y - 2 - \sqrt{3} = 4x - 2\sqrt{3} + 3$$

$$\Rightarrow 4y = 4x - 2\sqrt{3} + \sqrt{3} - 3 + 2 \Rightarrow \boxed{y = x - \frac{\sqrt{3}}{4} - \frac{1}{4}}$$

The equation of the line tangent to the curve $\theta = \pi/6$ for $r = 1-\cos\theta \Rightarrow$

$$\boxed{y = x - \frac{\sqrt{3}}{4} - \frac{1}{4}}$$

(B) Vertical tangent line, where $\frac{dx}{d\theta} = 0$

$$\therefore \quad \frac{dx}{d\theta} = r' \cos\theta - r \sin\theta = 0$$

$r = (1 - \cos\theta)$, $r' = \sin\theta$

$$\frac{dx}{d\theta} = \sin\theta\cos\theta - \sin\theta(1-\cos\theta) = 0$$

$$\frac{dx}{d\theta} = \sin\theta\cos\theta - \sin\theta + \sin\theta\cos\theta = 0$$

$$dx/d\theta = 2\sin\theta\cos\theta - \sin\theta = 0$$

$$dx/d\theta = \sin\theta(2\cos\theta - 1) = 0$$

$\therefore \sin\theta = 0$, $2\cos\theta - 1 = 0$

$\theta = 0, \pi, 2\pi$, $\cos\theta = 1/2 \Rightarrow \theta = \pi/3, 5\pi/3$

when $\theta = 0 \Rightarrow r = 1 - 1 = 0$

$\theta = 2\pi \Rightarrow r = 1 - 1 = 0$

$\theta = \pi/3 \Rightarrow r = 1 - 1/2 = 1/2$

$\theta = 5\pi/3 \Rightarrow r = 1 - 1/2 = 1/2$

$\theta = \pi \Rightarrow r = 1 - (-1) = 2$

∴ The tangent line is vertical

when $\theta = \pi, \pi/3, 5\pi/3$. which points are $(2, \pi), (1/2, \pi/3), (1/2, 5\pi/3)$.

(3)

$$F = \int_a^b P \cdot g \,(\text{length of the plate})\,(\text{depth})\, dy$$

$y = x^{1/3}$ and $y = -x^{1/3}$

$P = 1200 \text{ kg/m}^3$

$$F = \int_0^2 1200 \times 9.8 \,(\text{length})(\text{depth})\, dy$$

$$F = \int_0^2 1200 \times 9.8 \,(2)(8 - y^3)(2 - y)\, dy$$

$$F = 23520 \int_0^2 (8 - y^3)(2 - y)\, dy$$

$$F = 23520 \int_0^2 (16 - 2y^3 - 8y + y^4)\, dy$$

$$F = 23520 \left[16y - \frac{2y^4}{4} - \frac{8y^2}{2} + \frac{y^5}{5} \right]_0^2$$

$$F = 23520 \left[16(2) - \frac{(2)^4}{2} - 4(2)^2 + \frac{(2)^5}{5} \right]$$

$$F = 23520 \left[32 - 8 - 16 + \frac{32}{5} \right]$$

$$F = 23520 \left[8 + \frac{32}{5} \right] = 23520 \left[\frac{72}{5} \right]$$

$$\boxed{F = 338688 \ N}$$

(4) The approximating methods such as Trapezoidal Rule, Simpson's Rule and Taylor Series are helping to find complicated functions or improper integrals or infinite number of fractions and minimize the error occurs on approximation. We can find exact areas, volume, or Integrals, ect. But they have to be in particular shape or they have to be in particular condition. When people try to get larger number of data or finding a smallest pastic's data, they have to use approximation. Otherwise, the numbers might increase or decrease. For example, finding population, growth of bacteria.

However, approximate the polynomial or series is not easy as finding approximation of area of the curve. Taylor series helps to find those polynomial and series in more efficient way. Taylor series approximate the area by taking derivatives of function. So, we don't need diagram of the curve or of a function, which makes the Taylor series a far superior method.

(5) I liked the sequences and series because in computer science, sequences and series are one of the important topic to know. Most of the computer language's basic is sequence and series. For example, Python is an object-oriented programming language, which is using series as a programming code. Knowing sequence and series, how is it works and how it can be solve, are also interesting. This also helps us to improve critical thinking and logical thinking while finding order of sequence.

In addition, I liked the fluid mechanics because it is the most challeging part of calculus 2. It requires a lot of knowledge to understand that how fluid pressure works, which makes more

Sense why we constructing dams in certain ways.